

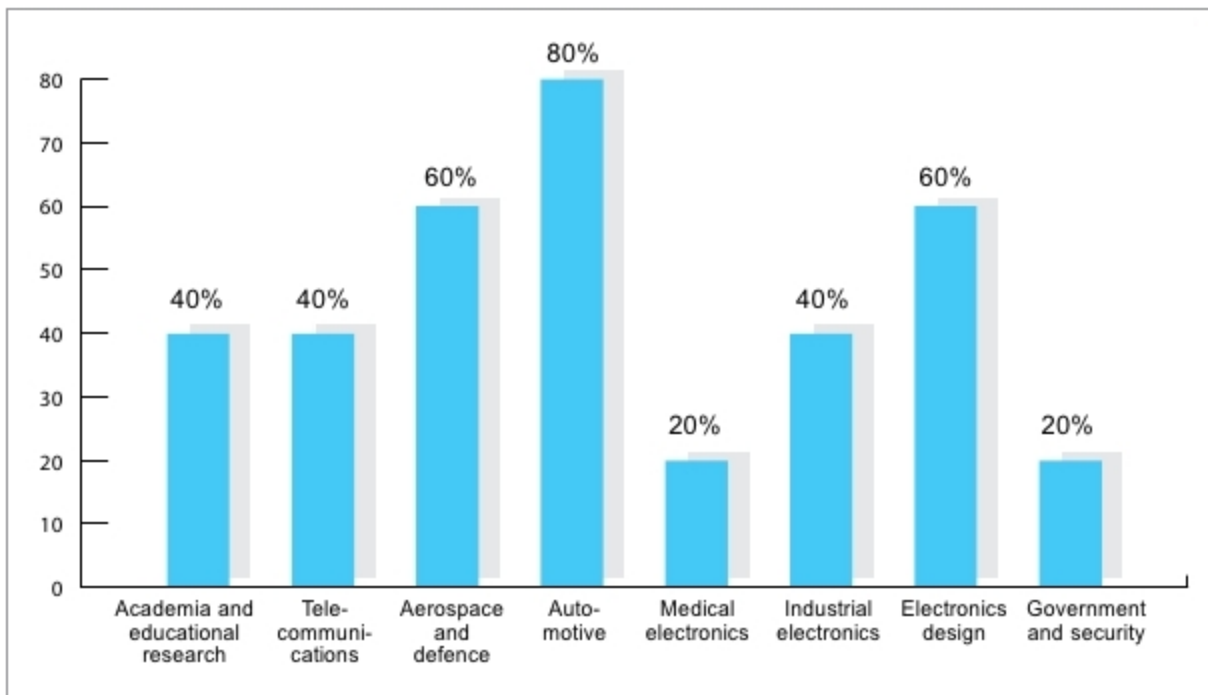


Fig. 2: Forecast on major demand-generating application sectors in India for T&M market

and domestic demand. Growth in this market is a result of the growth registered by key end-user segments like automotive, aerospace and defence, electronics design and more. Defence and other government verticals have been instrumental in driving the growth of this sector. Our experts forecast automotive, aerospace and defence as well as electronics design as the major demand-generating application sectors (Fig. 2).

The trend of automotive companies switching to automated testing is also driving up the demand for T&M equipment. Moreover, increasing electronic content in the vehicles for infotainment, connectivity and safety purposes, along with growing acceptance of green mobility (through e-vehicles) will also fuel the growth of the T&M sector. Besides, an increase in R&D activities across the electronics industry value chain is expected to further drive up demand. Considering the strong government push to make in India in the aerospace and defence electronics sector, the experts mentioned the bright future of the domestic defence manufacturing ecosystem and related opportunities for T&M.

India has vast potential in telecommunications and mobile manufacturing, too. With Make in India initiative creating a wave of optimism and opportunities, mobile manufacturing is set to gain further momentum. Therefore experts are looking

forward to the burgeoning opportunities in mobile handset manufacturing and allied verticals, like mobile services, cable TV and broadcasting.

Aforementioned government initiatives are likely to boost the power and communications sectors in India and eventually contribute to the growth of the T&M industry.

Demand for electricity across India has been growing at a rapid rate and is expected to grow further in the years to come. Consequently, power generation, transmission and distribution companies will have to invest in reliable, accurate and efficient equipment to protect their assets from failure. Therefore growth prospects of T&M equipment related to the industrial electronics sector are quite promising.

India is an emerging solar hub of the world, considering the kind of capacities that are coming up, with nearly 100GW of solar power to be added in the next five years. Solar power, again, may be a very important market for T&M products.

Besides demand-generating application sectors, the T&M market will be driven by some special trends as given below.

Support for predictive maintenance. Increasing awareness about predictive maintenance among manufacturers in India is driving the growth of the T&M industry.

Equipment with unique features. Demand from customers to address complex process issues is pushing

T&M companies to innovate continuously. This has resulted in the development of high-end devices with unique features.

Multi-functional and multi-domain testing. Multi-functional testing kits are the latest trend among decision makers of maintenance departments. Instead of having many different testing kits for individual applications, engineers have realised the benefits of using one kit with multi-functional testing capabilities. Such tools serve multi-domain applications as well.

Smart testing for smart devices. For testing managers and engineers, the challenge is to ensure quality at increasingly lower costs. To test their smart devices, organisations are transitioning from the status quo of rack-and-stack box instruments and closed turnkey ATE systems to smarter test systems that deliver connectivity and problem-solving capabilities.

Customised equipment. The current focus on customised test equipment is driving the overall market, as most of the low-end T&M equipment are now available with control options.

Quality ensures long-term success. According to a majority of experts, customers are willing to spend the extra penny for good quality. However, the definition of quality can be subjective and is often determined by the customer's unique perspective. Nevertheless, experts on the panel for this report believe that objective standards of quality lead to quantifiable benchmarks, as every customer segment wants its systems to function properly and reliably on a consistent basis.

Emerging trends

The need for a connected and safer world leveraging advancements in the Internet of Things (IoT), machine-to-machine (M2M) interaction, 5G and millimetre are the new technological trends that will lead to the growth of the T&M industry worldwide and in India, too.

Proliferation of smart devices has begun to influence the design and use of T&M equipment. Each smart product must go through a specific